Performed experimental precedent for generated complete reactions

Selected cases; complete net chemistry, not experimentally established electron-arrow sequences.

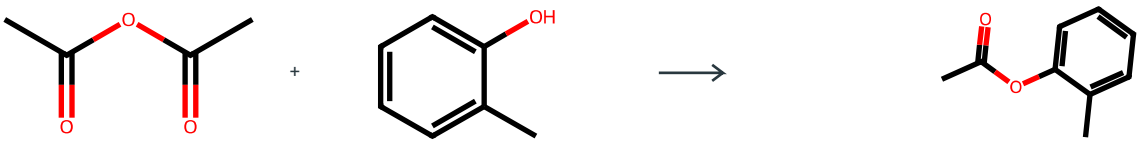
1. Tier A · US4645771:example25:hydroxybenzoyl-chloride

4-Hydroxybenzoic acid 11.04 g, thionyl chloride 80 mL and DMF 0.1 mL, reflux 2 h; evaporation and toluene coevaporation. Residue containing the named acid chloride was carried into the subsequent amidation.

2. Tier A · US8629137B2:intermediate39:step1

3,5-Difluorobenzoic acid (source reports 1.63 g, 10.00 mmol), thionyl chloride 20 mL, boiling 2 h; evaporate to dryness and coevaporate twice with toluene. Source reports 1.45 g, 82% of theoretical, as crude product used in Step 2.

3. Tier A · US5750050:step1.3:cresyl-acetate



o-Cresol 21.6 g/0.2 mol in aqueous NaOH (12 g/0.3 mol, water 100 mL), ice 300 g; acetic anhydride 30.6 g/0.3 mol, shake 1 min. Ether extraction and reduced-pressure distillation give 25 g, 83.3%; boiling point 66 C/1 mmHg.

Source/training overlap unknown; an experimentally documented transformation is not a tested whole route.